

Forecasting 3D Hand Motion with Spatial and Frequency Anisotropic Residual Diffusion

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Abstract. This submission is currently a structured scaffold for an ACVR 2026 / ECCV 2026 workshop full paper on FiHARD. The final abstract should briefly cover four elements: the assistive-computer-vision or human-robot-interaction motivation; the core method, centered on frequency-domain coarse forecasting and residual diffusion with hand-structured priors; the main empirical findings, which are still TODO; and the workshop relevance, such as anticipatory assistance, manipulation support, rehabilitation, or embodied interaction.

TODO: write a self-contained abstract in fully polished prose once the final method wording and experimental results are fixed. Do not include non-anonymous code links, author-identifying phrasing, or unsupported claims.

Keywords: 3D hand motion forecasting · assistive computer vision · diffusion models · human-robot interaction

1 Introduction

Anticipating hand motion is a useful capability in egocentric and interactive settings, where the hand is both the primary actuator [9] and a strong cue for upcoming action [11, 15]. In embodied assistance, teleoperation, and human-robot interaction, forecasting how the hand will move in the near future can inform intended manipulation and support anticipatory interaction [12, 14]. Such forecasts can also provide motion cues that may be useful in assistive monitoring settings. More broadly, the goal is not only to recognize what a user is doing now, but also to model how the motion is likely to unfold over the next few steps.

This forecasting problem is especially demanding for 3D hand motion. Compared with broader articulated motion, hand trajectories are more densely structured, more locally coupled, and more sensitive to fine-grained deformation associated with grasping or interaction. The future therefore contains both stable trend-like components and ambiguous residual variations: some aspects of the motion are predictable from context and recent dynamics, while others remain multi-modal at the level of individual articulators. A practical model must capture both aspects without sacrificing articulated consistency.

Existing approaches only partially address this combination of structure and uncertainty. Deterministic sequence predictors can capture dominant motion

trends effectively [8, 17, 19], and strong feed-forward baselines further reinforce this point [10], but such models tend to collapse multiple plausible futures into a single average prediction. Stochastic formulations can represent variability more explicitly [1, 4, 26]. More recent diffusion-based predictors pursue the same goal from a generative perspective [23, 24], yet generic generative formulations do not necessarily encode articulated hand structure or the fact that future uncertainty may be uneven across motion frequencies. This motivates a forecasting formulation that separates predictable trajectory trend from residual uncertainty while structuring the latter around hand-specific motion geometry.

We address this problem with FiHARD, a coarse-to-fine model for 3D hand-motion forecasting. The method first predicts a deterministic coarse future in the frequency domain, using a compact DCT representation to capture the dominant spectral structure of the trajectory. It then refines this prediction with a conditional diffusion model defined over future-frequency residuals. The residual process is organized by a spatio-spectral prior that combines articulated hand topology with frequency-dependent future modes, so that uncertainty is modeled along directions induced by articulated hand structure and future-frequency variation. In this way, FiHARD treats forecasting as the combination of coarse deterministic prediction and structured stochastic residual refinement. The full formulation is detailed in Sec. 3.

Our contributions are as follows:

- We introduce a frequency-domain coarse-to-residual formulation for 3D hand-motion forecasting, in which a deterministic predictor captures the dominant future trajectory and a stochastic model focuses on the remaining residual uncertainty.
- We develop a conditional diffusion model over future-frequency residuals, where sampling refines a coarse forecast rather than generating the entire trajectory from scratch.
- We define a spatio-spectral anisotropic residual prior that combines articulated hand topology and future-frequency modes to structure the residual uncertainty space.
- We evaluate the resulting model against stochastic and deterministic forecasting baselines on standard 3D hand-motion benchmarks, with quantitative results and ablations reported in Sec. 4.

2 Related Work

2.1 3D Human and Hand Motion Forecasting

Forecasting articulated motion has been studied extensively in the broader human-motion literature. Early and influential deterministic models relied on recurrent or feed-forward temporal prediction [19], while later work improved long-range structure through motion attention [17], graph reasoning [8], and stronger feed-forward sequence modeling [10]. In parallel, stochastic methods were introduced to represent the multi-modal nature of future motion through latent-variable

modeling [1, 26], diverse-hypothesis generation [16, 18], and probabilistic multi-modal forecasting [21]. These directions establish the main trade-off that also appears in hand forecasting: deterministic models often capture dominant trends well, whereas stochastic models are better suited to ambiguous futures.

Hand forecasting narrows this problem to a more densely articulated and manipulation-oriented signal. Compared with whole-body forecasting, hand motion contains stronger local couplings among nearby joints and sharper deformations associated with grasping or interaction. Recent egocentric hand-trajectory works study future hand motion as an anticipatory signal for interaction understanding [15], egocentric 3D hand trajectory forecasting [2, 11], diffusion-based hand prediction [24], and interaction-aware trajectory reasoning [5]. This paper considers that hand-specific setting and models future 3D hand motion through a deterministic coarse trend followed by stochastic residual refinement.

2.2 Frequency-Domain Motion Modeling

Frequency-domain representations provide a compact way to model smooth motion trends and long-range temporal structure. In particular, DCT-based forecasting has shown that a trajectory can often be predicted effectively from a low-dimensional spectral representation, reducing the burden on the predictor to model frame-level variation directly [10]. This idea is attractive for hand forecasting because a substantial portion of the predictable trajectory trend can be captured by low-frequency components, while higher-frequency variation is harder to model deterministically. Our method builds on this perspective by using a frequency-domain coarse predictor. Unlike methods that use frequency representations only as a compact deterministic forecasting space [10], FiHARD also defines residual stochastic refinement over future-frequency coefficients.

2.3 Diffusion Models for Motion Forecasting

Diffusion models provide a flexible framework for modeling structured uncertainty through iterative denoising [13, 20, 22]. They have been used for motion generation [25], motion completion for prediction [4], stochastic human motion forecasting [23], behavior-driven motion prediction [3], and nonisotropic diffusion for 3D motion prediction [7]. For forecasting, diffusion is especially appealing because it can generate multiple plausible futures conditioned on the observed past. However, these formulations are commonly introduced without an explicit hand-structured spatio-spectral prior. FiHARD differs by centering diffusion around a deterministic coarse forecast and by structuring the residual noise process through hand topology and future-frequency modes.

2.4 Hand-Centric Anticipation in Egocentric and Interactive Settings

Egocentric and interactive perception settings motivate forecasting models that anticipate how people will move before an action is complete. In egocentric vision, hand trajectories provide a direct cue about future interaction with objects

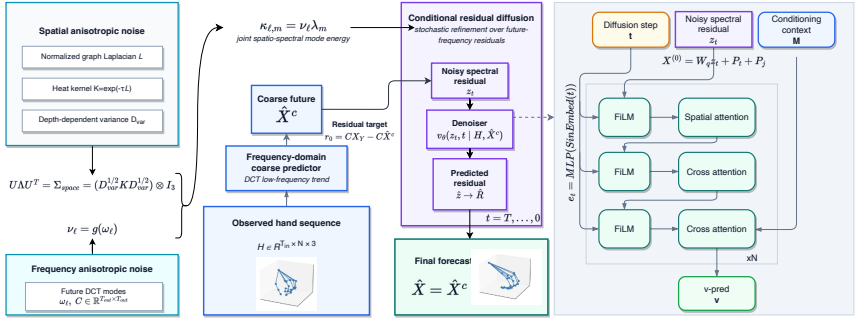


Fig. 1: Overview of FiHARD. The model first predicts a coarse future in the frequency domain, then refines it with conditional residual diffusion in a spatio-spectral space structured by articulated hand topology and future-frequency anisotropy.

and the surrounding environment [2, 5]. In human-robot collaboration, related work treats probabilistic motion prediction as a component for anticipatory assistance and safer interaction [14]. In teleoperation, uncertainty-aware intention prediction serves a related anticipatory role [12]. These settings motivate hand-motion forecasting models that are both stochastic and structurally constrained. Our work remains focused on 3D hand-motion forecasting, but adopts this anticipatory perspective by modeling future hand motion as a distribution over articulated residual trajectories.

3 Method

We formulate FiHARD as a coarse-to-fine model for 3D hand-motion forecasting. The method combines a deterministic frequency-domain predictor with a conditional diffusion model over residual future motion. Its central design choice is to represent uncertainty in a spatio-spectral residual space shaped jointly by articulated hand structure and frequency-dependent future motion modes. An overview of the full architecture is shown in Fig. 1.

3.1 Problem Formulation

Let

$$H \in \mathbb{R}^{T_{\text{in}} \times N \times 3}, \quad Y \in \mathbb{R}^{T_{\text{out}} \times N \times 3}$$

denote the observed and future 3D hand trajectories, where T_{in} and T_{out} are the observation and prediction horizons and N is the number of joints. We use the flattened representations

$$X_H \in \mathbb{R}^{T_{\text{in}} \times F}, \quad X_Y \in \mathbb{R}^{T_{\text{out}} \times F}, \quad F = 3N.$$

The forecasting problem is decomposed into two stages. First, a deterministic branch predicts a coarse future $\hat{X}^c$. Second, a stochastic refinement branch

models the residual motion that remains after this coarse prediction has been formed.

3.2 Coarse Frequency-Domain Forecasting

The coarse branch operates in a DCT representation of the full observation–prediction window. Let

$$T = T_{\text{in}} + T_{\text{out}}$$

and let $\tilde{X} \in \mathbb{R}^{T \times F}$ denote the temporally extended sequence obtained by appending a simple continuation of the observed motion to the input history. We project this sequence onto a low-frequency DCT basis,

$$Z_{\text{low}} = C_{\text{low}} \tilde{X},$$

where C_{low} contains the low-frequency rows of the orthonormal DCT matrix over the full window.

A sequence predictor then operates on this compact frequency representation and estimates the dominant spectral structure of the full trajectory. Drawing inspiration from SiMLPe [10], which provides a strong and simple frequency-domain forecasting baseline, we instantiate this predictor as a stack of temporal MLP layers over the DCT coefficients. The predicted low-frequency coefficients are mapped back to the time domain and restricted to the prediction horizon, producing the coarse future

$$\hat{X}^c \in \mathbb{R}^{T_{\text{out}} \times F}.$$

This stage captures the large-scale and relatively stable components of the future motion while leaving the remaining uncertainty to the residual model.

Since the stochastic branch acts on future-frequency residuals, we also define the DCT representation of the coarse future over the prediction horizon:

$$\hat{Y}_f^c = C \hat{X}^c,$$

where $C \in \mathbb{R}^{T_{\text{out}} \times T_{\text{out}}}$ is the orthonormal DCT matrix over the prediction horizon.

3.3 Spatial and Frequency Anisotropy

Spatial prior. The spatial prior is defined from the articulated hand topology. We consider a hand graph

$$G = (V, E),$$

where the nodes represent joints and the edges encode kinematic adjacency. To represent non-uniform residual variability across the hand, we assign depth-dependent variance weights to the joints. Let

$$D_{\text{var}} = \text{diag}(d_1, \dots, d_N)$$

be the diagonal matrix of jointwise variance weights. These weights act as a node-wise amplitude prior: more constrained joints receive smaller variance weights, whereas more mobile articulators receive larger ones.

To couple neighboring joints, we define a graph-smooth covariance factor from the hand topology. Let A be the adjacency matrix of the hand graph and D_g its degree matrix. We form the normalized graph Laplacian [6]

$$L = I - D_g^{-1/2} A D_g^{-1/2},$$

and the associated heat kernel

$$K = \exp(-\delta L),$$

where $\delta > 0$ controls the smoothing scale on the hand graph. This kernel is symmetric positive definite and diffuses covariance over the graph, so that nearby kinematic neighbors receive stronger cross-joint coupling than distant joints.

We combine graph smoothness and heteroskedastic joint variance at the node level as

$$\Sigma_{\text{node}} = D_{\text{var}}^{1/2} K D_{\text{var}}^{1/2},$$

then lift this covariance to 3D coordinates through

$$\Sigma_{\text{space}} = \Sigma_{\text{node}} \otimes I_3.$$

The heat-kernel factor determines how covariance spreads over the hand graph, while D_{var} controls the marginal uncertainty scale at each joint. This construction yields spatial residual modes that are simultaneously graph-smooth and non-uniform across articulators.

We write the eigendecomposition of the resulting spatial covariance as

$$\Sigma_{\text{space}} = U A U^\top,$$

where U is the spatial spectral basis and

$$A = \text{diag}(\lambda_1, \dots, \lambda_F)$$

contains the spatial mode energies. This basis defines the articulated residual directions used by the diffusion model.

Frequency anisotropy. The stochastic branch also distinguishes between future motion modes of different frequency. Let

$$\omega_\ell, \quad \ell = 1, \dots, T_{\text{out}},$$

index the future DCT modes. We assign a nonnegative frequency weight to each mode,

$$\nu_\ell = g(\omega_\ell),$$

where $g(\cdot)$ is a monotone spectral weighting function. In the present model, this weighting plays the role of a temporal or frequency anisotropy term: lower- and higher-frequency future modes need not be corrupted at the same rate.

The spatial and frequency components are combined into a spatio-spectral anisotropic schedule. For frequency mode ℓ and spatial mode m , we define the joint mode energy

$$\kappa_{\ell,m} = \nu_{\ell} \lambda_m.$$

These joint energies determine how strongly each residual mode is noised during diffusion. As a result, uncertainty is not represented isotropically: it is structured both across articulated hand directions and across future frequency modes. Intuitively, this formulation changes the corruption schedule by no longer treating all residual coefficients equally. Residual modes that vary more rapidly over the forecast horizon receive larger frequency weights and are therefore noised more aggressively. On the spatial side, modes with larger prior residual variability—typically those emphasizing more mobile articulated regions while remaining compatible with the hand graph—also receive faster corruption. Conversely, smoother and more topology-consistent residual modes retain signal longer during the forward process.

3.4 Conditional Residual Diffusion

The stochastic branch models residual motion in the future frequency domain. Let

$$r_0 = CX_Y - C\hat{X}^c$$

denote the clean future-frequency residual over the full hand-coordinate representation. Rather than diffusing this residual directly in the original coefficient coordinates, we express it in the spatial spectral basis induced by the covariance prior:

$$z_0 = r_0 U.$$

Here the DCT matrix C acts along the temporal prediction axis, while the spatial basis U acts along the F -dimensional hand-coordinate axis. Therefore $z_0 \in \mathbb{R}^{T_{\text{out}} \times F}$ remains indexed jointly by future frequency and spatial residual mode.

Given a diffusion timestep t , the noised residual state is defined by mode-dependent perturbation in this spectral space:

$$z_t = \sqrt{\bar{\alpha}_t} \odot z_0 + \sqrt{1 - \bar{\alpha}_t} \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I),$$

where the cumulative attenuation factors are applied elementwise over the spatio-spectral modes and depend on the anisotropic schedule introduced above. Hence, residual directions associated with different articulated structures or different future frequency modes are corrupted at different rates.

The denoiser is conditioned on the observed history and on the coarse future. Its role is to predict the standard v target associated with the pair (z_0, ϵ) at timestep t , rather than reconstructing the clean residual directly. This choice yields a conditional residual diffusion model whose stochastic component is centered around the coarse forecast instead of replacing it.

After denoising, the residual is mapped back from the spatial spectral basis as $\hat{r} = \hat{z}U^\top$, inverted from the future DCT representation, and added to the coarse trajectory. The final forecast is thus

$$\hat{X} = \hat{X}^c + \hat{R},$$

where $\hat{R}$ is the decoded residual trajectory, expressed in the same trajectory coordinate system as the coarse branch.

Proposition 1 (Validity of the spatio-spectral forward process). *For the spatio-spectral forward process introduced above, the following properties hold:*

1. the forward noising process is admissible, with nonnegative variance in every spatio-spectral mode;
2. the induced process in the residual coordinates is linear-Gaussian;
3. the process is Markov, with mode-wise one-step transitions; and
4. the process is compatible with standard v -prediction.

This proposition formalizes the fact that the proposed diffusion process is a valid anisotropic forward model over future-frequency residuals. A proof is provided in the supplementary appendix.

3.5 Training and Inference

FiHARD is trained in two phases. The first phase optimizes the coarse forecasting branch so that it captures the dominant predictive structure of future hand motion. The second phase trains the residual diffusion branch while allowing the coarse predictor to remain jointly refinable after the initial warmup. This preserves consistency between the deterministic stage and the stochastic refinement stage, rather than treating the coarse forecast as a fixed external module.

At inference time, the model first predicts the coarse future trajectory, then forms its future-frequency representation, samples a residual correction through iterative denoising in spatio-spectral space, and finally reconstructs the future hand trajectory by adding the decoded residual to the coarse prediction. Optimization details, schedule parameters, and architecture hyperparameters are deferred to the experimental section.

4 Experiments

We evaluate FiHARD on three datasets: Assembly, H2O, and BigHands. Following the current evaluation logs, we report Average Displacement Error (ADE), Final Displacement Error (FDE), Average Pairwise Distance (APD), CMD, and FID for all compared models. Lower ADE, FDE, CMD, and FID indicate better forecasting accuracy and distribution matching, while higher APD indicates greater sample diversity.

Table 1 reports Assembly results grouped by action, arranged as pairs of subtables across three rows. Table 2 places the H2O and BigHands results side

Table 1: Assembly results grouped by action. Best values in each column are bold; second-best values are underlined.

(a) <i>attempt</i>						(b) <i>pick</i>					
Model	ADE ↓	FDE ↓	APD ↑	CMD ↓	FID ↓	Model	ADE ↓	FDE ↓	APD ↑	CMD ↓	FID ↓
FiHARD	0.698	0.966	4.886	<u>0.008</u>	0.036	FiHARD	TODO	TODO	TODO	TODO	TODO
comusion	<u>0.719</u>	<u>0.979</u>	9.365	0.005	0.296	comusion	<u>0.872</u>	1.140	<u>11.384</u>	0.006	0.306
belfusion	1.644	1.769	5.688	0.041	0.399	belfusion	1.793	1.875	6.981	0.030	0.088
dlow_cvae	0.884	1.102	7.634	0.013	<u>0.165</u>	dlow_cvae	1.001	<u>1.201</u>	8.834	0.011	0.007
humanmac	0.888	1.138	<u>9.635</u>	0.010	0.259	humanmac	1.101	1.346	11.604	<u>0.008</u>	0.183
skeletoondiffusion	1.022	1.305	7.149	0.012	0.357	skeletoondiffusion	1.269	1.580	10.756	0.006	1.610
gsps	0.778	1.070	10.461	0.011	0.501	gsps	0.867	1.140	9.611	0.010	<u>0.030</u>

(c) <i>put</i>						(d) <i>remove</i>					
Model	ADE ↓	FDE ↓	APD ↑	CMD ↓	FID ↓	Model	ADE ↓	FDE ↓	APD ↑	CMD ↓	FID ↓
FiHARD	TODO	TODO	TODO	TODO	TODO	FiHARD	<u>0.834</u>	1.160	5.844	0.009	<u>0.015</u>
comusion	<u>0.907</u>	<u>1.183</u>	11.615	0.002	0.169	comusion	0.802	1.084	9.438	0.004	0.000
belfusion	1.767	1.824	7.042	0.030	0.129	belfusion	1.678	1.736	6.292	0.034	0.163
dlow_cvae	1.052	1.260	9.371	0.011	0.000	dlow_cvae	0.999	1.235	8.654	0.016	0.281
humanmac	1.043	1.310	<u>10.945</u>	0.008	<u>0.017</u>	humanmac	0.965	1.209	<u>9.909</u>	0.008	0.047
skeletoondiffusion	1.385	1.752	9.227	<u>0.006</u>	1.923	skeletoondiffusion	1.320	1.663	8.681	<u>0.007</u>	1.522
gsps	0.881	1.165	9.026	0.012	<u>0.017</u>	gsps	0.844	<u>1.148</u>	10.145	0.010	0.150

(e) <i>rotate</i>						(f) <i>unscrew</i>					
Model	ADE ↓	FDE ↓	APD ↑	CMD ↓	FID ↓	Model	ADE ↓	FDE ↓	APD ↑	CMD ↓	FID ↓
FiHARD	0.918	<u>1.255</u>	6.203	<u>0.011</u>	0.143	FiHARD	0.747	<u>0.968</u>	3.136	<u>0.009</u>	0.015
comusion	0.967	1.280	13.578	0.006	0.880	comusion	<u>0.752</u>	0.910	<u>9.535</u>	0.004	0.578
belfusion	1.895	1.951	6.562	0.046	0.500	belfusion	1.454	1.471	5.830	0.038	0.402
dlow_cvae	1.145	1.356	9.511	0.017	<u>0.270</u>	dlow_cvae	0.935	0.983	7.539	0.020	<u>0.284</u>
humanmac	1.201	1.503	<u>13.266</u>	<u>0.011</u>	0.718	humanmac	0.879	1.035	8.294	0.018	0.410
skeletoondiffusion	1.566	1.971	9.531	0.014	2.110	skeletoondiffusion	1.147	1.343	6.894	0.019	1.589
gsps	<u>0.948</u>	1.247	9.643	0.018	0.399	gsps	0.794	0.976	10.192	0.020	0.740

by side. We keep explicit TODO markers for the Assembly actions whose FiHARD runs are still missing from the current results file.

Overall, the Assembly results show that FiHARD is consistently strong on the action subsets for which all required runs are already available. It achieves the best ADE on *attempt*, *rotate*, and *unscrew*, and it also obtains the best FDE on *attempt* and the best FID on *attempt*, *rotate*, and *unscrew*. The *remove* subset is more competitive, where comusion slightly improves ADE, FDE, CMD, and FID, but FiHARD remains the second-best ADE model and retains low distributional error. On H2O, FiHARD remains competitive but is not the best model under ADE and FDE: comusion attains the strongest reconstruction accuracy, while FiHARD is second-best in ADE and maintains a substantially lower FID than several older baselines. On BigHands, FiHARD becomes one of the strongest models overall, ranking second on ADE, tying for best on CMD and FID, and remaining competitive on FDE, while the highest APD values are obtained by other methods.

Figure 2 complements the tables by visualizing the diversity-quality tradeoff through APD and ADE. The main pattern is that the methods with the largest diversity are not necessarily the strongest overall generators: for instance, gsps

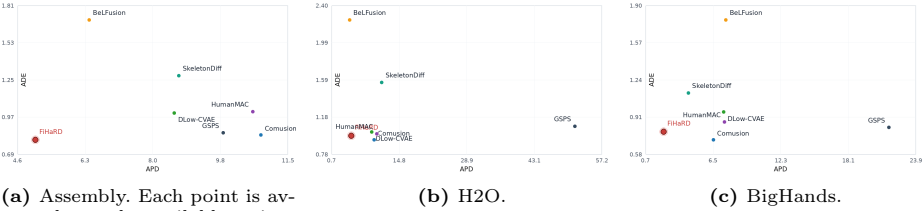


Fig. 2: Diversity-quality tradeoff measured as APD versus ADE on the three benchmarks. Higher APD indicates greater diversity, while lower ADE indicates better forecast quality. Across the three datasets, the plots show that the largest diversity values do not automatically correspond to the best forecast quality, and that FiHARD tends to remain in a favorable low-error region without requiring extreme APD.

Table 2: Cross-dataset results. Best values in each column are bold; second-best values are underlined.

(a) H2O						(b) BigHands					
Model	ADE ↓	FDE ↓	APD ↑	CMD ↓	FID ↓	Model	ADE ↓	FDE ↓	APD ↑	CMD ↓	FID ↓
FiHARD	<u>0.979</u>	1.387	4.837	0.010	<u>0.070</u>	FiHARD	<u>0.778</u>	1.254	2.220	0.003	0.000
comusion	0.933	1.252	9.600	0.009	0.000	comusion	0.705	1.075	6.505	0.003	0.000
belfusion	2.244	2.467	4.483	0.078	0.956	belfusion	1.770	1.781	<u>7.574</u>	0.048	0.000
dlo_w_cvae	1.020	1.322	9.118	0.010	0.000	dlo_w_cvae	0.952	1.327	7.390	0.005	0.000
humanmac	1.000	<u>1.300</u>	10.184	0.007	0.000	humanmac	0.863	<u>1.169</u>	7.452	<u>0.004</u>	0.000
skeletondiffusion	1.561	1.770	<u>11.195</u>	<u>0.008</u>	0.396	skeletondiffusion	1.122	1.545	4.354	<u>0.004</u>	0.000
gspgs	1.083	1.558	51.519	0.020	28.044	gspgs	0.815	1.202	21.578	0.011	<u>7.499</u>

tends to push farther toward high APD, but this often comes with weaker forecast quality. In contrast, FiHARD consistently remains in a favorable low-error region, especially on Assembly and BigHands, without relying on extreme APD. This suggests that the proposed model preserves multimodality while keeping the samples close to the target motion distribution.

These trends are consistent with the theoretical formulation introduced in Sec. 3. Since FiHARD performs diffusion in a residual space aligned with articulated hand topology and frequency-dependent motion modes, its stochastic refinement is not imposed in an unconstrained isotropic latent space. Empirically, this structure appears to improve sample consistency with the target distribution: across several Assembly actions and on BigHands, FiHARD combines strong ADE/FDE with particularly competitive FID and CMD values. We therefore interpret the results as evidence that the spatio-spectral formulation improves not only forecasting accuracy, but also the quality-consistency tradeoff of the generated futures.

5 Limitations

TODO: discuss dataset and annotation constraints, especially if the final benchmarks do not include full visual or object context.

TODO: discuss modeling constraints such as fixed horizons, dependence on consistent joint indexing, and assumptions made by the current residual diffusion formulation.

TODO: discuss deployment limitations for assistive or interactive settings, including latency, robustness to occlusion, and out-of-distribution behavior if relevant.

TODO: keep this section specific and balanced, without overstating future impact.

6 Conclusion

TODO: summarize the final method and empirical takeaways in concise form.

TODO: restate the ACVR relevance in a grounded way, such as anticipatory assistance, hand-aware interaction, or manipulation support.

TODO: avoid exaggerated claims and avoid mentioning manuscript provenance or supplementary material plans here.

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